HuddleAI

Product Requirements Document

Version 2.0 | April 2026

Product	HuddleAI
Version	2.0
Status	Active Development
URL	huddleai.app
Stack	React, TypeScript, Firebase, OpenAI GPT-4, Google Cloud Speech-to-Text
Last Updated	April 2026

1. Overview

HuddleAI is an AI-powered meeting intelligence layer. It automatically processes, transcribes, and summarizes meetings — then delivers those insights directly into the tools teams already use.

HuddleAI is NOT a project management tool. It does not compete with Jira, Linear, Notion, or Asana. It integrates with them. The product's value is in making meeting intelligence invisible — action items appear in Jira, summaries land in Notion, and teams barely notice a new tool has been added.

2. Problem Statement

Teams of all sizes lose significant time and context after meetings. Action items get lost, decisions go undocumented, and follow-up is inconsistent. Existing solutions either:

- Require teams to adopt entirely new project management workflows (high friction, low adoption)
- Produce summaries that live in a silo — disconnected from where work actually happens
- Are built exclusively for large enterprise teams, ignoring individual and casual users

The result is that meeting intelligence never makes it into the tools people depend on daily.

3. Product Vision

Make meeting outcomes automatically flow into every tool a team already uses — with zero extra effort from users. And for individuals, make every meeting instantly searchable, summarized, and shareable.

Design Principle

For every feature decision, ask: does this make meeting intelligence flow better into tools teams already use, or does it make the solo experience simpler and more useful? If neither, don't build it.

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4. Target Audience

Solo / Casual Users — Free Forever

- Individuals who want to capture and summarize their own meetings
- Non-technical users — friends, freelancers, students, small business owners
- No team setup required — start immediately after signup
- Gets: transcription, plain text summary, action items list, export. No integrations.
- Upgrades to team features whenever ready — never prompted

Team Users (Free) — Friends & Small Projects

- Solo users who want to share meeting summaries with a few friends or teammates
- Small projects, side projects, early-stage teams with no budget
- Gets: everything in Solo, plus shared team workspace and teammate invites. No integrations.

Team Users (Paid) — Professional Teams

- Tech-forward engineering and product teams using Jira, Linear, Notion, Google Workspace
- Want full automation — meetings processed, action items pushed to Jira, summaries in Notion
- Gets: everything in Team Free, plus all integrations, auto-ingest, admin controls, SSO

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5. Pricing Tiers

Plan	What's Included
Solo	Upload MP3/MP4 (up to 1GB). AI transcription and plain text summary. Action

Free forever	items list. Export/copy summary. English only. No integrations.
Team (Free) Free forever	Everything in Solo. Shared team workspace. Invite teammates — no credit card required. No integrations. Ideal for friends and small projects.
Team (Paid) Paid / per team	Everything in Team Free. Full integration suite: Slack, Jira, Linear, Asana, Notion, Confluence, Google Drive. Auto-ingest from Google Meet and Zoom. Searchable transcript history. Recurring meeting context. Admin controls and SSO.

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6. Out of Scope

The following are explicitly out of scope and will not be built:

- Scrum boards or kanban views
- Internal task assignment and status tracking within HuddleAI
- Sprint planning or backlog management
- Any feature that positions HuddleAI as a standalone project management tool
- Multi-language support (English only for MVP)

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7. Technical Constraints (MVP)

- Supported file formats: MP3 (audio) and MP4 (video)
- Maximum file size: 1GB
- Language support: English only
- Transcription: Google Cloud Speech-to-Text
- AI summarization: OpenAI GPT-4

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8. Onboarding & User Flow

Every new user starts as a Solo user by default. There is no team setup prompt, no onboarding wizard, no friction. The first experience is:

- Sign up → land on dashboard → upload a meeting → get a summary
- No team creation required to get value
- Team features (shared workspace, integrations) are available in settings whenever the user is ready
- Upgrading to a team plan is self-serve — user invites teammates at their own pace

The goal is that a non-technical user (a friend, a freelancer, a student) can sign up and get a useful summary within 5 minutes, without ever thinking about teams or integrations.

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9. Action Items — Reframe

The existing action items feature is valuable but needs to be repositioned. The current implementation has CRUD-style task management built in, which conflicts with the product vision.

What to Keep

- AI extraction of action items from meeting transcripts
- Display of extracted action items in meeting detail view
- Simple action items list across all meetings — not a kanban, just a clean unified view
- Push action items to external tools (Jira, Linear, Asana, Slack) for paid team users

What to Remove or Reframe

- Internal status tracking (mark complete, in progress, etc.) — this is project management
- Internal assignment within HuddleAI — assignees belong in Jira, not HuddleAI
- Any UI that makes action items feel like a task manager

The reframe: action items are meeting outputs, not tasks. The primary CTA on every action item is 'Send to Jira' or 'Send to Linear' — not 'Mark complete.'

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10. Current State (v0)

The following features are already built and deployed at huddleai-a812c.web.app (migrating to huddleai.app):

- Meeting upload (MP4 video files up to 1GB)
- Automatic audio extraction via FFmpeg
- Transcription with speaker diarization (Google Cloud Speech-to-Text)
- AI-generated summaries, action items, topics, decisions (GPT-4)
- Slack integration — manual and automatic meeting notifications
- Team workspaces with role-based access control
- Firebase Auth, Firestore, Cloud Functions infrastructure
- Custom domain: huddleai.app (Cloudflare + Firebase Hosting)

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11. Phased Roadmap

Phase	Focus	Key Deliverables
Phase 1	**Focus & Clean Up ✅**	COMPLETE. Removed task management UI, reframed action items, added MP3 support, fixed routing, built userStage utility.
Phase 2	**Onboarding & Workspace**	Onboarding fork (solo vs team). Workspace switcher in nav. Personal workspace always exists. Multi-team support. Data ownership rules. Team deletion flow.
Phase 3	**Auto-Ingest**	Google Calendar + Meet (transcript-first, recording fallback). Zoom (Phase 3.1). Microsoft Teams (Phase 3.2). Manual upload always available.
Phase 4	**Integration Layer**	Jira v1 (manual trigger, field mapping UI). Linear and Asana (Phase 4.1+). Notion/Confluence summary sync. Google Drive archiving.
Phase 5	**Intelligence Layer**	Full-text search via Algolia or Typesense. Recurring meeting context. Team-level analytics. Weekly digest.

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12. Technical Stack

Frontend	React, TypeScript, Tailwind CSS, shadcn/ui
Backend	Firebase Cloud Functions (Node.js, TypeScript)
Database	Firestore
Storage	Firebase Storage
Auth	Firebase Authentication
AI / ML	OpenAI GPT-4, Google Cloud Speech-to-Text
Hosting	Firebase Hosting + Cloudflare (huddleai.app)
Search (Phase 4)	Algolia or Typesense (Firestore full-text not suitable)
Region	us-central1

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13. Success Metrics

Solo Tier

- Time from signup to first processed meeting (target: under 5 minutes)
- % of solo users who process at least 3 meetings in first 30 days

Team Adoption

- % of team users who connect at least one integration within 30 days (target: >60%)
- % of meetings processed via auto-ingest vs. manual upload (target: >70% post-Phase 2)

Retention

- Weekly active users at 30 / 60 / 90 days
- Teams with 2+ integrations connected (strongest retention signal)

Upgrade

- % of solo users who upgrade to a team plan within 90 days

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14. Engineering Instructions for Phase 1

The following instructions are strict directives for the AI-assisted development environment (Antigravity). Follow them in order. Do not skip steps or reorder them. Each task has a clear definition of done.

Task 1 — Remove /team-setup Redirect [Day 1, P0]

This is the first thing to fix. It is blocking the entire solo user vision.

What to do

- In App.tsx, find the post-signup redirect that routes new users to /team-setup
- Replace it with a redirect to /dashboard
- Ensure /team-setup still exists as a route (accessible from Settings) but is never shown automatically
- Remove any onboarding wizard or modal that appears on first login

Definition of done

- New user signs up → lands on /dashboard with upload CTA
- /team-setup is accessible via Settings but never triggered automatically
- No regression for existing users who already have teams

Task 2 — Remove Internal Task Management UI [P0]

Remove all UI that makes HuddleAI feel like a task manager. Do not remove the data — only the UI.

What to remove

- Scrum board component and its route
- 'Mark Complete', 'In Progress', 'To Do' status controls on action items
- Internal assignee field on action items (within HuddleAI — not the extracted assignee text)
- Any kanban or board view

What NOT to remove

- The action items list view — keep it
- AI-extracted assignee names displayed as text — keep them
- Action items displayed in meeting detail view — keep them

Definition of done

- No status tracking UI exists anywhere in the app
- Action items are displayed as a read-only list with extracted context
- No regression on meeting processing or summary display

Task 3 — Reframe Action Items as Meeting Outputs [P0]

Change the primary CTA on every action item from internal tracking to external push. In Phase 1, these CTAs are placeholders — the actual integrations are built in Phase 3.

What to build

- Each action item gets two placeholder buttons: 'Send to Jira' and 'Send to Linear'
- Clicking either shows a toast: 'Jira integration coming soon — upgrade to Team (Paid)'
- Add 'Copy to clipboard' as a functional action on each action item
- Add 'Copy all action items' button at the top of the action items list

Definition of done

- Every action item has Copy, Send to Jira (placeholder), Send to Linear (placeholder)
- No 'Mark Complete' or status UI anywhere
- Copy to clipboard works correctly

Task 4 — Add MP3 Support [P0]

The current pipeline only handles MP4. Extend it to accept MP3 audio files.

What to change

- Firebase Storage rules: add audio/mpeg to allowed MIME types alongside video/*
- Upload UI: update file input validation to accept .mp3 in addition to .mp4
- Upload UI: update help text to say 'Supports MP3 and MP4, up to 1GB'
- Cloud Function (processMeetingUpload): detect MP3 input and handle accordingly — MP3 does not need audio extraction via FFmpeg, pass directly to Speech-to-Text
- Test: upload a real MP3 file end-to-end and confirm transcript + summary are generated

Definition of done

- MP3 files upload without error
- MP3 files are transcribed and summarized correctly
- MP4 files continue to work without regression
- Files over 1GB are rejected with a clear error message

Task 5 — Solo Onboarding Flow [P0]

The first-time experience for a new user should be: sign up → see dashboard → upload a meeting. Nothing else.

What to build

- Empty state on /dashboard for new users: headline 'Upload your first meeting', subtext 'Get an AI summary in minutes', large Upload button
- Remove any team-related prompts, modals, or banners from the new user flow
- 'Invite teammates' option lives in Settings → Team, not in onboarding
- No changes to the signup form itself — email + password is fine

Definition of done

- New user lands on /dashboard after signup with clear upload CTA
- No team setup prompt appears at any point in the default flow
- Existing users with teams are not affected

General Rules for All Phase 1 Work

Do not build

- Email delivery — this is Phase 2. Do not add any email infrastructure in Phase 1.
- Any new integrations — Jira, Linear, Slack changes are Phase 3. Placeholders only.
- Auto-ingest — this is Phase 2. Manual upload remains the only input method in Phase 1.

Code standards

- All changes in TypeScript — no plain JS
- No new Firestore collections without explicit approval
- Do not modify the Cloud Function processing pipeline beyond MP3 support

- All UI changes must be mobile-responsive
- Deploy to huddleai.app after each task is complete and tested

Testing requirements

- Each task must be manually tested before marking done
- Test both solo user flow AND existing team user flow for every change
- No regressions on meeting upload, processing, or summary display are acceptable

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15. User Flow — Solo to Team

Every user starts as Solo regardless of intent. The product earns trust through the core value loop before asking for anything. Team features are discovered, not forced.

Stage 1 — Solo Empty (zero meetings, no team)

This is every new user's first experience. The entire screen is focused on one action.

- Dashboard shows: large headline 'Upload your first meeting', subtext 'Get an AI summary in minutes', single Upload button
- No stats, no team UI, no Slack prompt, no integration banners
- Header: HuddleAI logo + Upload Meeting button only
- No mention of teams, plans, or integrations anywhere on screen

Stage 2 — Solo Active (has meetings, no team)

User has uploaded at least one meeting. They've seen the core value. Dashboard becomes useful.

- Dashboard shows: recent meetings list, basic stats (total meetings, total duration)
- Action items list accessible from nav — clean read-only list, no task management
- No team UI, no Slack prompt, no integration banners
- Subtle 'Invite a teammate' link in Settings only — never surfaced on the main dashboard
- Header: logo + Upload Meeting button

Stage 3 — Team Free (has team, free plan)

User has invited at least one teammate. Shared workspace is now relevant.

- Dashboard label changes to 'Team Dashboard'

- Shared meetings visible to all team members
- Team Members section visible in dashboard
- Slack Integration prompt now surfaces — it's relevant at this stage
- Jira/Linear CTAs on action items show 'Coming soon — upgrade to Team (Paid)'
- Stats reflect the whole team's activity

Stage 4 — Team Paid (has team, paid plan)

User has upgraded. Integrations are unlocked. This is the full product experience.

- All integrations active: Slack, Jira, Linear, Asana, Notion, Confluence, Google Drive
- 'Send to Jira' and 'Send to Linear' CTAs are functional, not placeholders
- Auto-ingest settings visible (Phase 2+)
- Admin controls visible to team owner
- Full stats and analytics dashboard

userStage Utility — Engineering Spec

The dashboard must read a userStage value and render the correct view. Build a userStage utility function that returns one of four string constants based on the following logic:

- SOLO_EMPTY — user has 0 meetings AND is not a member of any team
- SOLO_ACTIVE — user has 1+ meetings AND is not a member of any team
- TEAM_FREE — user is a member of a team AND team plan is free
- TEAM_PAID — user is a member of a team AND team plan is paid

The dashboard component imports userStage and renders the appropriate view. No other component should contain stage-branching logic — it all flows from this single utility.

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16. Engineering Instructions for Phase 2

IMPORTANT CONTEXT FOR ANTIGRAVITY: This is PRD v2.0. Everything built in Phase 1 and Sprint 2A remains valid and will not be changed. The work on userStage, action items reframe, MP3 support, and routing fixes is complete and correct. Phase 2 is a new layer on top of that foundation — not a replacement.

The Google Meet auto-ingest work (previously called Sprint 2B) is NOT cancelled. It has been pushed to Phase 3. The full spec from PRD v1.6.2 remains correct and will be the Phase 3 engineering brief. Do not start that work until Phase 2 is complete.

Phase 2 — Onboarding Fork + Workspace Switcher + Data Model

Phase 2 is a significant UI and data model change. It introduces the concept of workspaces — Personal and Team — as distinct contexts a user can switch between. Nothing from Phase 1 is undone. This is additive.

Sprint 2A — Data Model: Workspace Context [P0, Do First]

Before any UI changes, the data model needs to support workspace context. Every meeting must know whether it belongs to a personal workspace or a team workspace. This is foundational — all UI work depends on it.

What to build

- Add workspaceType field to every meeting document in Firestore: 'personal' | 'team'
- Add workspaceId field: null for personal meetings, teamId for team meetings
- Update MeetingUploadData interface to include workspaceType and workspaceId
- Update processMeetingUpload Cloud Function to write workspaceType and workspaceId based on upload context
- Update getMeetingsByUser query to filter by workspaceId — personal workspace shows only personal meetings, team workspace shows only that team's meetings
- Migrate existing meetings: any meeting with a teamId gets workspaceType='team', any meeting without gets workspaceType='personal'

Data ownership rules — implement these exactly

- Upload in personal workspace → workspaceType='personal', workspaceId=null, visible only to uploader
- Upload in team workspace → workspaceType='team', workspaceId=teamId, visible to all team members immediately
- Meetings cannot be moved between workspaces after upload — context is set at upload time, immutable
- If a user leaves a team: their access to that team's meetings is revoked. workspaceType and workspaceId on the meeting documents are unchanged — the meeting stays with the team
- If a user deletes a team they are the sole member of: prompt the user to move meetings to personal workspace OR delete permanently. If move: update workspaceType='personal', workspaceId=null on all affected meetings. If delete: permanently remove all meeting documents and associated storage files

Definition of done

- All new meetings correctly tagged with workspaceType and workspaceId at upload time
- All existing meetings migrated with correct workspace tags
- Dashboard only shows meetings belonging to the currently active workspace
- No regression on meeting upload, processing, or display

Sprint 2B — Onboarding Fork [P0]

The current onboarding sends all new users directly to the dashboard with no context. Phase 2 introduces a one-time choice between Personal and Team as the first screen after signup.

The fork screen — what to build

- Show this screen once only — immediately after signup, before landing on the dashboard
- Two cards side by side: 'Just me' and 'With a team'
- Reassurance line below the cards: 'Not sure? Start solo — switching to a team takes seconds from the nav'
- This screen is never shown again after the user makes a choice — store onboardingComplete: true in Firestore user document
- Users who signed up before this feature ships are treated as having chosen 'Just me' — do not show them the fork retrospectively

If user picks 'Just me'

- Show a confirmation screen: 'You're all set' headline, 'Upload your first meeting' subtext, Upload Meeting CTA button
- Show a one-time hint card: 'Working with others later? Switch to a team anytime from the workspace menu in the nav.'
- This hint appears once on the confirmation screen only — never again on the dashboard
- Proceed to SOLO_EMPTY dashboard

If user picks 'With a team'

- Show a second screen with two options: 'Create a team' and 'Join a team'
- 'Create a team' → team name input → create team → land on TEAM_FREE dashboard in that team's workspace
- 'Join a team' → invite code or link input → join team → land on TEAM_FREE dashboard in that team's workspace
- Back button available to return to the fork screen

Invited users — special case

- If a new user signs up via an invite link, skip the fork screen entirely
- They are already joining a team — land them directly in that team's workspace
- Their personal workspace still exists and is accessible from the workspace switcher

Definition of done

- New organic signup sees the fork screen exactly once
- Choosing 'Just me' leads to confirmation screen with hint, then SOLO_EMPTY dashboard
- Choosing 'With a team' leads to create/join screen, then correct team dashboard
- Invite-link signups bypass the fork entirely
- Returning users never see the fork again
- Existing users are unaffected

Sprint 2C — Workspace Switcher + Nav [P0]

The nav needs a workspace switcher so users can move between Personal and any teams they belong to. This is the primary mechanism for solo → team transition and back.

What to build

- Workspace switcher component in the top-left nav, next to the HuddleAI logo
- Switcher shows the name of the currently active workspace with a caret/chevron
- Clicking opens a dropdown with: Personal workspace at top, then all teams the user belongs to, then a quiet 'Create or join a team' option at the bottom
- Active workspace has a checkmark indicator
- Each workspace entry shows an icon (initials-based), workspace name, and member count for teams
- Switching workspace updates the entire dashboard context — meetings list, stats, action items all reflect the selected workspace
- Nav links (Meetings, Action items, Settings) remain the same regardless of workspace
- 'Upload Meeting' button context-aware — uploads go to the currently active workspace

Solo user with no teams

- Switcher shows 'Personal' with caret
- Dropdown shows only: Personal workspace (checked) + 'Create or join a team' at bottom
- No team section header shown if user has no teams — don't show an empty section
- The 'Create or join a team' option is small and quiet — never a banner or prominent CTA

User with multiple teams

- All teams listed in the dropdown under a 'Teams' section header
- No limit on number of teams — scroll if needed
- Creating a new team adds it to the list immediately

Definition of done

- Workspace switcher present in nav for all users
- Switching workspace correctly filters all dashboard content to that workspace
- Solo user never sees team prompts on the main dashboard — only in the dropdown
- Upload correctly assigns workspaceType based on active workspace at time of upload
- All existing functionality works correctly in both personal and team workspace contexts

Sprint 2D — Team Management [P1]

Basic team management flows needed to support the workspace model.

What to build

- Create team flow: team name input, optional logo/color, create → lands in new team workspace
- Join team flow: invite code or link input → validate → join → land in team workspace
- Invite member flow: Settings → Team → invite by email → sends invite link
- Leave team: Settings → Team → Leave team → confirmation prompt → revokes access to team meetings
- Delete team (sole member only): Settings → Team → Delete team → prompt to move meetings to personal or delete permanently
- Team owner concept: first person to create the team is the owner. Owner can invite, remove members, and delete the team. Members can only leave.

Definition of done

- Full create → invite → join flow works end to end
- Leaving a team correctly revokes meeting access
- Deleting a team as sole member correctly handles meeting migration or deletion
- Team owner vs member permissions enforced throughout

General Rules for All Phase 2 Work

Do not build in Phase 2

- Google Meet auto-ingest — this is Phase 3. Full spec in Section 16 of PRD v1.6.2.
- Jira, Linear, or any external integrations — Phase 4
- Pricing or billing infrastructure — not in scope yet
- Any changes to the AI processing pipeline

Code standards

- All changes in TypeScript — no plain JS
- No new Firestore collections without explicit approval — extend existing collections
- All new UI must be mobile-responsive
- All new UI must respect userStage — no hardcoded assumptions
- Test locally first — deploy only when sprint is complete and locally validated

Testing requirements

- Test all three signup paths: organic solo, organic team, invite link
- Test workspace switching with multiple teams
- Test upload in personal workspace vs team workspace — confirm correct workspaceType written
- Test team deletion flow — both move to personal and delete permanently paths
- Full regression on meeting upload, processing, and display after every sprint deploy

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17. Decision Record

This section documents significant product decisions and the reasoning behind them. These are final decisions — they should not be revisited without a strong new reason.

Decision: Phase 2 Reprioritised — Onboarding and Workspace Before Auto-Ingest

What was decided

Google Meet auto-ingest (previously Phase 2) has been pushed to Phase 3. A new Phase 2 covering onboarding fork, workspace switcher, and data model has been inserted before it.

Why

- Building auto-ingest on top of a nav and data model that was about to change would mean rebuilding integration work — wasteful and risky
- The workspace model (personal vs team) needs to exist before auto-ingest, because auto-ingest needs to know which workspace to deliver recordings into
- The onboarding fork is the first thing every new user sees — getting it right before growing the user base matters more than any specific feature
- All previous auto-ingest specs remain valid — they are just sequenced correctly now

Decision: Personal Workspace Always Exists

What was decided

Every user has a Personal workspace that can never be deleted or taken away. Teams are additive. Joining or leaving a team has no effect on a user's personal meetings or personal workspace.

Why

- Eliminates the 'dead end' feeling of being locked into a team or solo identity
- Reduces fear of trying team features — users know their personal data is safe
- Mirrors the mental model of tools users already know: Notion, Slack, Linear all have personal + team workspaces

Decision: Meeting Workspace Context is Immutable

What was decided

A meeting's workspace context (personal or team) is set at upload time and cannot be changed afterwards. There is no 'move to team' or 'move to personal' feature for existing meetings.

Why

- Simplifies the data model significantly — no complex migration logic needed
- Sets clear expectations: if you want a meeting in a team, upload it in the team workspace
- Prevents accidental data exposure — a meeting can't inadvertently be moved to a team context
- Exception: team deletion by sole member — this is the only case where meetings can be migrated, and it requires explicit user action

Decision: Email Delivery Removed from Roadmap

What was decided

Email delivery of meeting summaries has been permanently removed. In-app display and Slack cover delivery better with less engineering cost. Will only be reconsidered if users explicitly request it at scale.

Decision: Transcript-First Processing for Google Meet (Phase 3)

Google Meet auto-ingest will attempt to pull Google's native transcript first (available on Workspace Business Standard+), falling back to recording-based processing. This is faster, cheaper, and better for the highest-value user segment. Full spec in PRD v1.6.2 Section 16.

Decision: No Standalone Kanban or Task Management

Documented from v1.0. HuddleAI does not build internal task management. Action items are meeting outputs pushed to external tools. The primary CTA on every action item is 'Send to Jira' or 'Send to Linear', not 'Mark complete.'